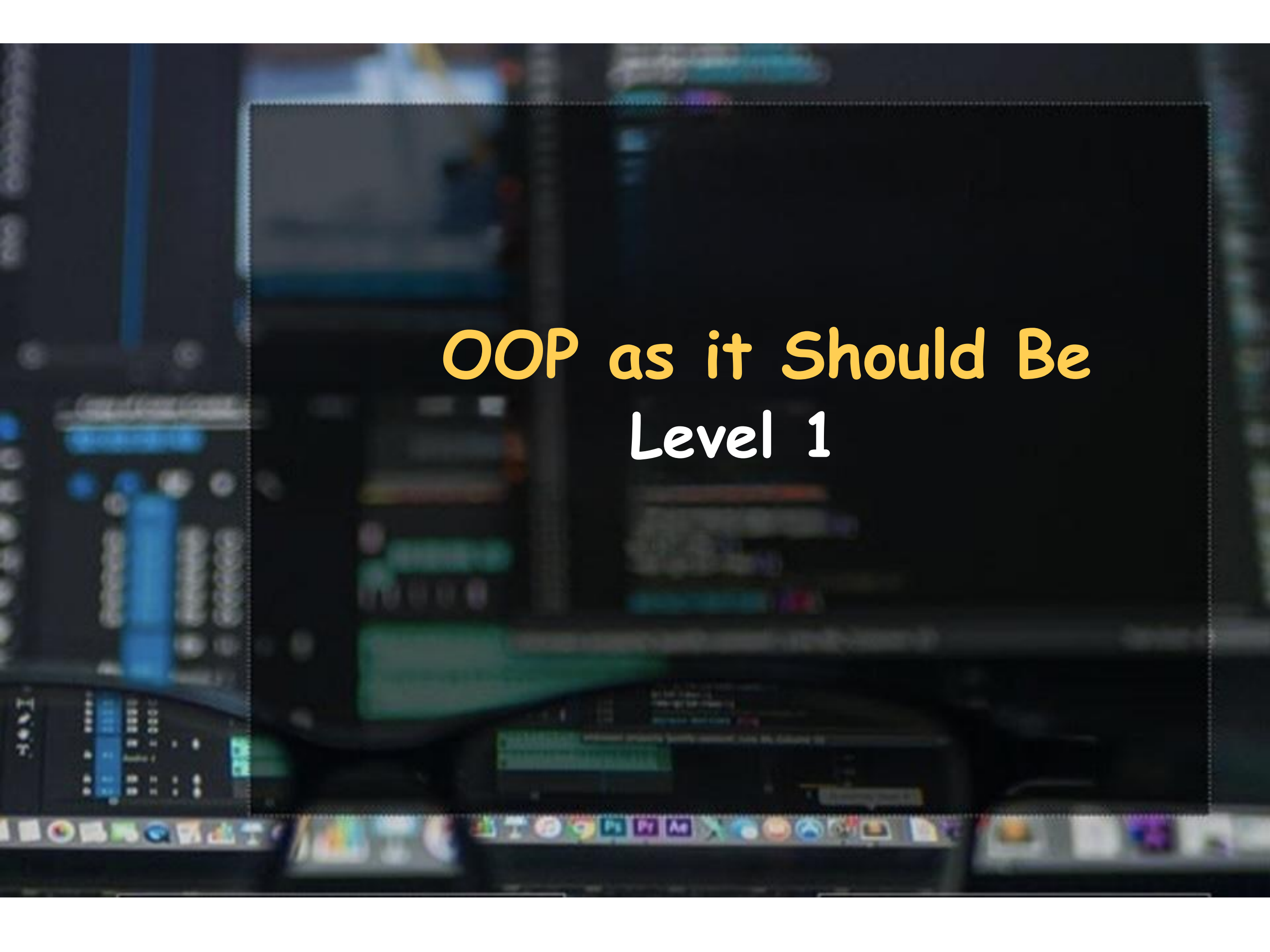




# OOP as it Should Be Level 1

The background of the slide features a complex, abstract pattern of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in motion, which adds a sense of activity and human presence to the technical theme.

# OOP as it Should Be

Level 1

## Properties Set and Get

## Definition

This example explains the concept of Setters and Getters in C++ using a class that represents a person with a first name and a last name.

## Explanation

- ◆ Uses Encapsulation to hide data and prevent direct access
- ◆ Data is accessed only through specific functions (*Set / Get*)
- ◆ Improves security, control, and code organization

# Components of clsPerson class

## 1 Private Data Members

-  \_FirstName : stores the first name
-  \_LastName : stores the last name

## 2 Setter Functions

-  setFirstName(string FirstName)
-  setLastName(string LastName)
-  Used to assign values to private variables

## 3 Getter Functions

-  FirstName()
-  LastName()
-  Used to read values without modifying them directly

## 4 FullName Function

-  FullName()
-  Combines first name and last name into one string

# Example

- ◆ Create an object from the class
- ◆ Assign values using setters
- ◆ Display values using getters

```
clsPerson Person1;
```

```
Person1.setFirstName("Sara");
```

```
Person1.setLastName("Alterkawi");
```

## Steps in main()

- 1 Create object Person1
- 2 Set first name
- 3 Set last name
- 4 Print first name
- 5 Print last name
- 6 Print full name



## Expected Output

First Name: Sara

Last Name: Alterkawi

Full Name: Sara Alterkawi



## Note

You cannot access `_FirstName` and `_LastName` directly because they are private.



# Comparison



## With Set/Get

- ✓ More secure
- ✓ Better control
- ✓ Cleaner code



## Without Set/Get

- ✗ Exposed data
- ✗ Harder to maintain

## **Warning**

Avoid making data members public unless necessary—it breaks encapsulation.

# Summary

- ✓ Set = write data
- ✓ Get = read data
- ✓ private protects data
- ✓ Core principle of Object-Oriented Programming

Thank you 😊